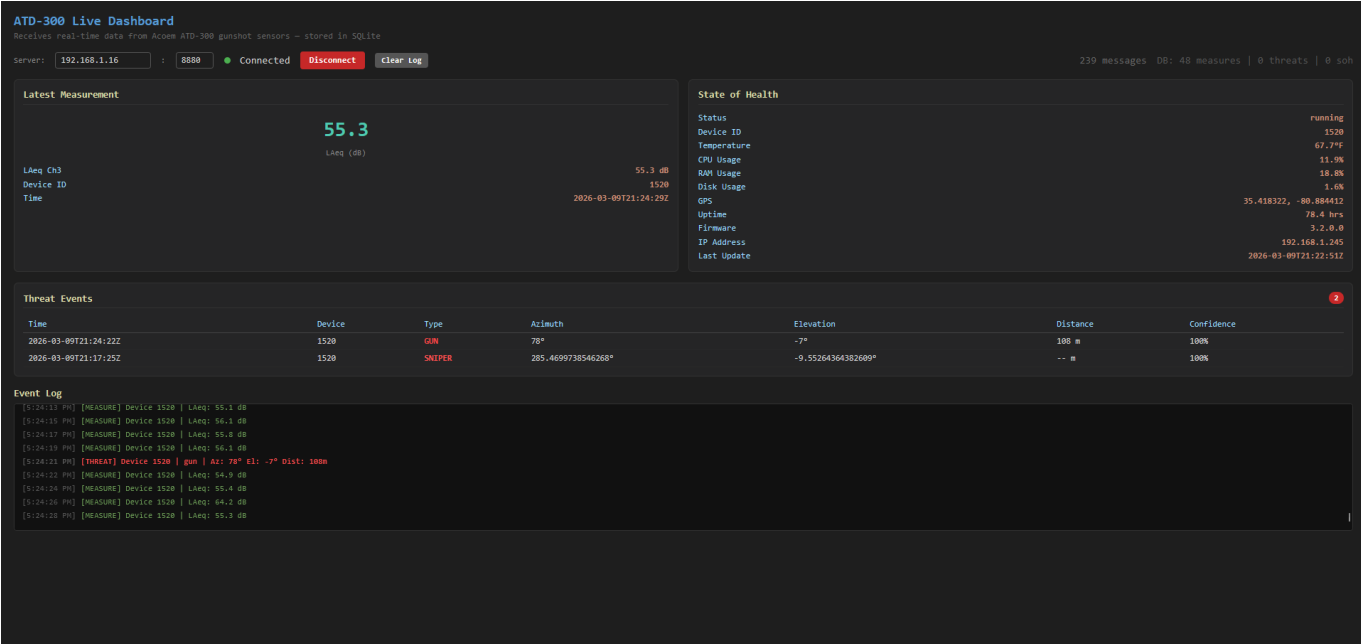
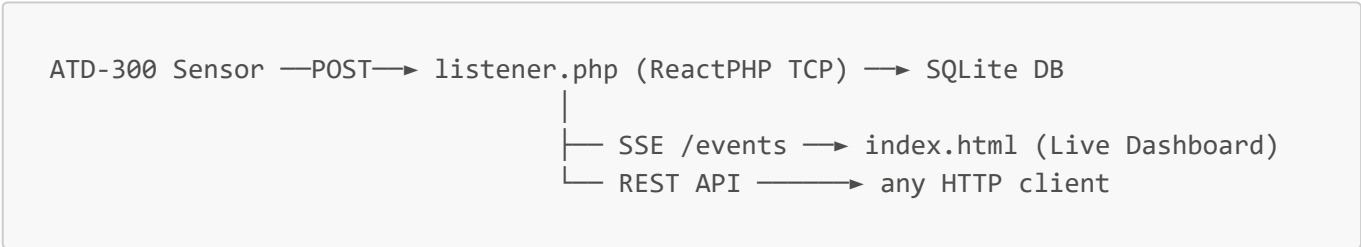


# ATD-300 Listener

A real-time TCP listener and dashboard for **Acoem ATD-300 gunshot detection sensors**. The server receives JSON data from sensors over TCP, stores it in a local SQLite database, and streams live updates to a browser-based dashboard via Server-Sent Events (SSE).



## Architecture



## Features

- **Async TCP server** powered by [ReactPHP](#) — handles sensor connections and browser clients on a single port
- **SQLite storage** with WAL mode for concurrent read/write
- **Live dashboard** ([index.html](#)) with real-time updates via SSE
- **Three data types** parsed and stored:
  - **Measures** — LAeq noise levels (dB)
  - **State of Health (SOH)** — device status, temperature, CPU/RAM/disk usage, GPS, firmware, uptime
  - **Threats** — gunshot detections with type, probability, azimuth, elevation, and distance
- **REST API** for querying stored data
- **Raw listener** ([raw\\_listener.php](#)) for low-level debugging of sensor traffic

## Requirements

- PHP 8.0+ with the `sqlite3` extension
- [Composer](#)

## Installation

```
git clone <repo-url> atd-listener
cd atd-listener
composer install
```

## Usage

### Start the listener

```
php listener.php
```

The server starts on **port 8880** and prints:

```
=== ATD-300 Listener on port 8880 ===
Sensor:   POST /measure | /soh | /threat
Dashboard: GET  /events (SSE)
API:      GET  /api/threats | /api/measures | /api/soh | /api/stats
Database: db/atd300.sqlite3
Waiting for data...
```

Alternatively, use the included helper script:

```
./go.sh          # runs listener.php in the background
```

### Open the dashboard

Open `index.html` in a browser. Enter the ATD-300 pod host/IP and port (defaults to `192.168.1.16:8880`), then click **Connect**. The dashboard will:

1. Load historical data from the API
2. Subscribe to the SSE stream for live updates
3. Display the latest measurement, SOH status, and a scrollable threat event table

### Raw debugging

To see exactly what bytes the sensor sends (useful for protocol debugging):

```
php raw_listener.php
```

This prints raw data and hex dumps for every incoming connection on port 8880.

## API Endpoints

| Method | Path                   | Description                                         |
|--------|------------------------|-----------------------------------------------------|
| GET    | /                      | Health check — returns <b>ATD-300 Listener OK</b>   |
| GET    | /events                | SSE stream of live sensor events                    |
| GET    | /api/threats?limit=50  | Recent threat detections (JSON)                     |
| GET    | /api/measures?limit=50 | Recent noise measurements (JSON)                    |
| GET    | /api/soh               | Latest state-of-health record (JSON)                |
| GET    | /api/stats             | Total counts for measures, threats, and SOH records |
| POST   | /measure               | Ingest a measurement from a sensor                  |
| POST   | /soh                   | Ingest a state-of-health report from a sensor       |
| POST   | /threat                | Ingest a threat detection from a sensor             |

## Database

Data is stored in `db/atd300.sqlite3` with three tables:

- measures** — `device_id`, `laeq`, `laeq_chan3`, `sensor_datetime`, `received_at`, `raw_json`
- soh** — `device_id`, `status`, `temp`, `cpu_usage`, `ram_usage`, `disk_usage`, `latitude`, `longitude`, `uptime`, `fw_version`, `ip`, `orientation`, `sensor_datetime`, `received_at`, `raw_json`
- threats** — `device_id`, `threat_type`, `probability`, `azimuth`, `elevation`, `distance`, `fw_version`, `nn_version`, `sensor_datetime`, `received_at`, `raw_json`

The database directory and tables are auto-created on first run. Schema mismatches are detected and tables are recreated automatically.

## Project Structure

```
|— listener.php      # Main async TCP server (ReactPHP)
|— raw_listener.php  # Raw TCP debug listener
|— index.html        # Live browser dashboard
|— composer.json     # PHP dependencies (react/socket)
|— go.sh             # Convenience startup script
|— db/               # SQLite database (auto-created)
|   └─ atd300.sqlite3
|— vendor/           # Composer dependencies
```

## Acoem ATD-300 Setup

Login to configure and setup the ATD-300 for testing.

Go to the Settings page and set the Server notification settings to match your server:



Status

General

Settings

System

Modes

User

Log

Cadence

Server notification settings

(Websocket)

Device identifier :

1520

Server address :

192.168.1.16

Server port :

8880

Update

VMS Alarms - Generic Event

Alarm for threat

HTTP (AXIS, Wisenet Wave...) ▾

All threats (No Authentication) :

Gun Shot alarm :

Sniper alarm :

Firecracker alarm :

VMS Authentication

false ▾

Update

Then go to the Modes tab and send a Gun Shot event of any king to simulate a gunshot. Alternatively you can put it in test mode or generate real gunshot events.

License

See individual dependency licenses in [vendor/](#).